

Summary of Series Tests

(1) **n-Term Test.** Given any series $\sum_{n=1}^{\infty} a_n$.

- If $\lim_{n \rightarrow \infty} a_n = 0$, the test is inconclusive.
 - If $\lim_{n \rightarrow \infty} a_n \neq 0$, the series diverges.
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(2) **Geometric Series.** $\sum_{n=1}^{\infty} ar^{n-1}$

- If $|r| < 1$, the series converges to $a/(1-r)$.
 - If $|r| \geq 1$, the series diverges.
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(3) **p-Series.** $\sum_{n=1}^{\infty} \frac{1}{n^p}$

- If $p > 1$, the series converges.
 - If $p \leq 1$, the series diverges.
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(4) **Integral Test.** If there exists a positive, continuous, decreasing function f on $[N, \infty)$ such that $a_n = f(n)$ for all $n \geq N$, then

- $\int_N^{\infty} f(x)dx$ and $\sum_{n=1}^{\infty} a_n$ both converge or both diverge.
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(5) **Direct Comparison Test.** For $\sum_{n=1}^{\infty} a_n$ with nonnegative terms, compare with a known series $\sum_{n=1}^{\infty} b_n$.

- If $a_n \leq b_n$ for all $n \geq N$ and $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.
 - If $a_n \geq b_n$ for all $n \geq N$ and $\sum_{n=1}^{\infty} b_n$ diverges, then $\sum_{n=1}^{\infty} a_n$ diverges.
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(6) **Limit Comparison Test.** Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be series with positive terms, evaluate $L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$.

- If $L > 0$, then $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ both converge or both diverge.
 - If $L = 0$ and $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.
 - If $L = \infty$ and $\sum_{n=1}^{\infty} b_n$ diverges, then $\sum_{n=1}^{\infty} a_n$ diverges.
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(7) **Ratio Test.** For any series $\sum_{n=1}^{\infty} a_n$ with nonzero terms, let $\rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$.

- If $0 \leq \rho < 1$, the series converges absolutely.
 - If $\rho = 1$, the test is inconclusive.
 - If $\rho > 1$ or $\rho = \infty$, the series diverges.
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(8) **Root Test.** For any series $\sum_{n=1}^{\infty} a_n$, let $\rho = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$.

- If $0 \leq \rho < 1$, the series converges absolutely.
 - If $\rho = 1$, the test is inconclusive.
 - If $\rho > 1$ or $\rho = \infty$, the series diverges.
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(9) **Alternating Series Test.** $\sum_{n=1}^{\infty} (-1)^{n+1} b_n$ or $\sum_{n=1}^{\infty} (-1)^n b_n$

- If $b_n \geq 0$ for all $n \geq 1$, the sequence $\{b_n\}$ is eventually decreasing (that is there is an integer N such that $b_n \geq b_{n+1}$ for all $n \geq N$) and $b_n \rightarrow 0$, then the series converges.
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Remark. (i) Absolute convergence implies convergence.

- (ii) If $\left| \frac{a_{n+1}}{a_n} \right|$ or $\sqrt[n]{|a_n|}$ does not tend to a limit as $n \rightarrow \infty$ (they are divergent, but not divergent to ∞), then the Ratio Test and Root Test tell us nothing.